PROJECT DELIVERABLE

Talaqqi

Bridging Tradition & Technology for Transformative Qur'an Learning

Use Case Diagram, Use Case Descriptions, and Analysis Class Diagram

Group Members Information

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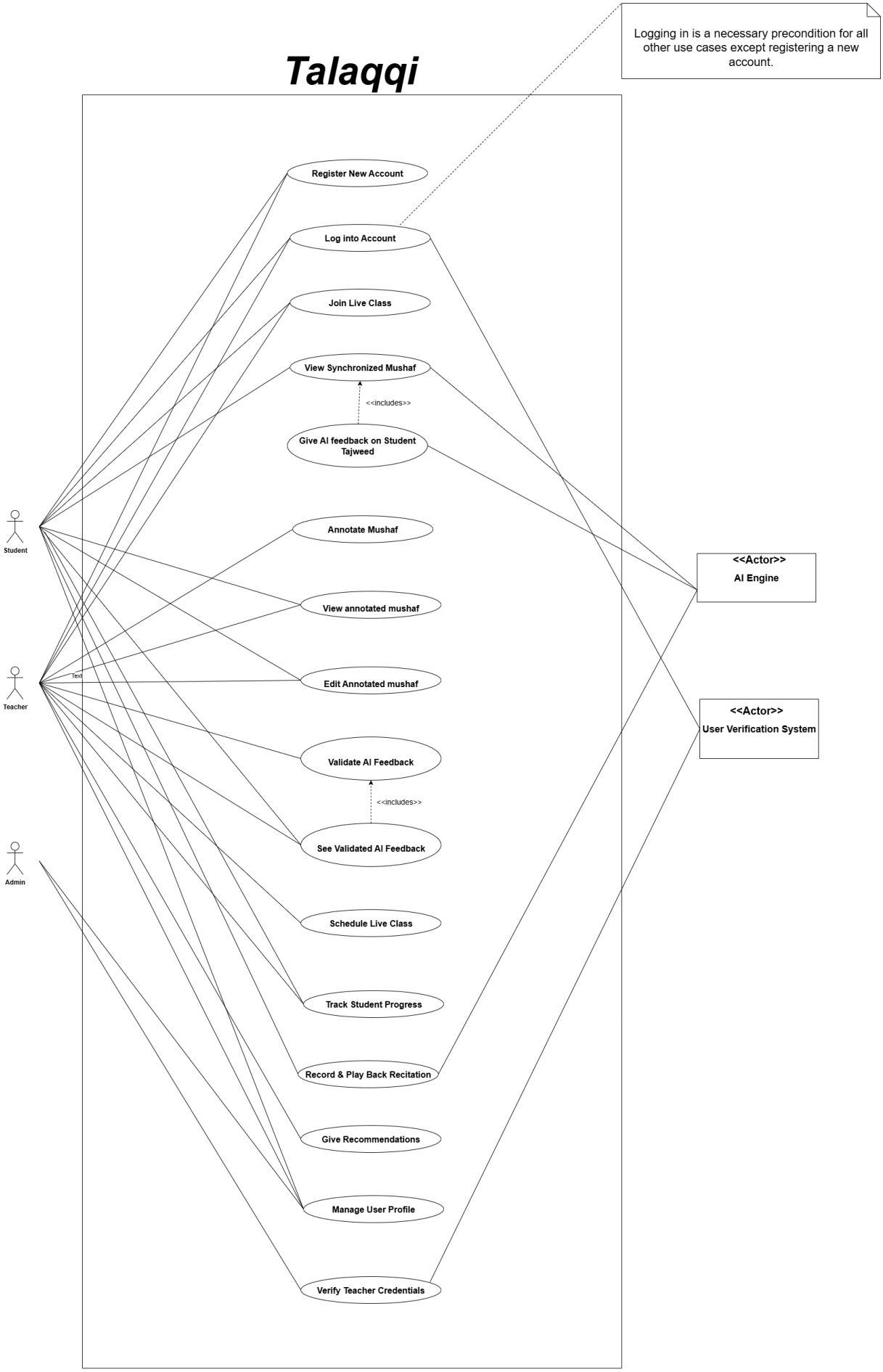
1. Use Case Diagram



2. Use Case Descriptions

2.1 UC-01: Register New User

Identifier	UC-01
Name	Register User
Purpose	Allow a Student, Teacher, or Admin to create a new account on the platform.
Priority	High
Actors	Student, Teacher, Admin, Authentication Service
Pre-conditions	The user must not already be registered. The Authentication Service must be available.
Post-conditions	A new account is created, and the user can proceed to login.
Dependencies	None

2.1.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
User selects the Register option.	Displays the registration form.
User enters required details (e.g., name, email, password).	Validates fields for completeness and format.
User submits the form.	Authentication Service verifies details and confirms registration.
—	Creates the user account and shows success confirmation.

2.1.2 Alternative Course of Action

Actor Action	System Response
User enters already registered email or username.	Displays error message: "Email or username already exists."
User enters invalid email format or weak password.	Displays specific validation error and prompts correction.

Exceptions:

- System is unavailable.
- Authentication Service is down.

2.2 UC-02: Login / Authenticate User

Identifier	UC-02
Name	Login / Authenticate User
Purpose	Enable users to securely log in to the platform with authentication.
Priority	High
Actors	Student, Teacher, Admin, Authentication Service
Pre-conditions	User must already be registered.
Post-conditions	User is logged into the system and redirected to their dashboard.
Dependencies	Register User

2.2.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
User selects the Login option.	Displays login form.
User enters username/email and password.	Validates input and sends credentials to Authentication Service.
—	Authentication Service verifies credentials.
—	Grants access and redirects to dashboard.

2.2.2 Alternative Course of Action

Actor Action	System Response
User enters invalid password or username/email.	Displays "Invalid credentials" message.
Unverified teacher attempts login.	Denies access and prompts for verification with message: "Account awaiting verification."

Exceptions:

- Authentication Service unavailable.
- System database inaccessible.

2.3 UC-03: Schedule Live Class

Identifier	UC-03
Name	Schedule Live Class
Purpose	Allow Teachers to schedule live classes for students.
Priority	High
Actors	Teacher
Pre-conditions	Teacher must be logged in and verified.
Post-conditions	Live class is scheduled and notifications sent to students.
Dependencies	Login / Authenticate User

2.3.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Teacher selects Schedule Class option.	Displays scheduling form.
Teacher enters date, time, and topic.	Validates details for conflicts and format.
Teacher submits schedule.	Saves class schedule and sends notifications to students.

2.3.2 Alternative Course of Action

Actor Action	System Response
Teacher enters invalid date/time (e.g., past date or conflicting schedule).	Displays error message: "Invalid or conflicting schedule. Please correct details."

Exceptions:

- System unavailable.
- Network error prevents scheduling.

2.4 UC-04: Join Live Class

Identifier	UC-04
Name	Join Live Class
Purpose	Enable Students and Teachers to join scheduled live classes.
Priority	High
Actors	Student, Teacher
Pre-conditions	A class must be scheduled. User must be logged in.
Post-conditions	User successfully joins the live session.
Dependencies	Schedule Live Class

2.4.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
User selects Join Class option.	Checks for scheduled class and user eligibility.
User confirms attendance.	Connects user to the live session.

2.4.2 Alternative Course of Action

Actor Action	System Response
User attempts to join without registration or login.	Denies access and prompts login with message: "Please log in to join the class."

Exceptions:

- Network connectivity issues.
- Class server unavailable.

2.5 UC-05: Synchronized Mushaf Viewing

Identifier	UC-05
Name	Synchronized Mushaf Viewing
Purpose	Allow Students and Teachers to view synchronized Mushaf during class.
Priority	High
Actors	Student, Teacher
Pre-conditions	User must be in a live class session.
Post-conditions	User views the Mushaf page synchronized by the teacher.
Dependencies	Join Live Class

2.5.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Teacher selects a page/verse in Mushaf.	Updates and synchronizes view for all participants.
Student views synchronized content.	Ensures correct alignment across users.

2.5.2 Alternative Course of Action

Actor Action	System Response
Student attempts to view Mushaf outside session.	Denies access with message: "Mushaf access requires an active class session."

Exceptions:

- System synchronization failure.
- Connectivity issues disrupt view.

2.6 UC-06: Annotate Mushaf

Identifier	UC-06
Name	Annotate Mushaf
Purpose	Enable Teachers to add annotations and notes to Mushaf for students.
Priority	Medium
Actors	Teacher, Student (viewer)
Pre-conditions	Teacher must be in session. Student must be logged in.
Post-conditions	Annotations appear for students in real time.
Dependencies	Synchronized Mushaf Viewing

2.6.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Teacher selects annotation tool.	Enables annotation interface.
Teacher adds notes/highlights to Mushaf.	Displays annotations to students in real time.

2.6.2 Alternative Course of Action

Actor Action	System Response
Teacher tries to annotate without an active session.	Denies access with message: "Annotation requires an active class session."

Exceptions:

- Annotation tool malfunction.
- Network delay in updates.

2.7 UC-07: AI Tajweed Analysis

Identifier	UC-07
Name	AI Tajweed Analysis
Purpose	Provide students with AI-based recitation feedback.
Priority	High
Actors	Student, Teacher, AI Engine
Pre-conditions	Student must record recitation.
Post-conditions	Provides AI-generated tajweed feedback.
Dependencies	Record
Playback Recitation	

2.7.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Student records recitation.	Forwards audio to AI Engine.
—	AI Engine analyzes tajweed and returns feedback.
—	Displays feedback to student and teacher.

2.7.2 Alternative Course of Action

Actor Action	System Response
Student provides unclear recitation.	Requests re-recording with message: "Audio unclear, please re-record."

Exceptions:

- AI Engine unavailable.
- Audio file corrupted.

2.8 UC-08: Edit Annotated Mushaf

Identifier	UC-08
Name	Edit Annotated Mushaf
Purpose	Allow Teachers to edit existing annotations on the Mushaf.
Priority	Medium
Actors	Teacher
Pre-conditions	Teacher must be logged in and have existing annotations.
Post-conditions	Annotations are updated and visible to students.
Dependencies	Annotate Mushaf

2.8.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Teacher selects existing annotation to edit.	Displays annotation editing interface.
Teacher modifies notes/highlights and submits changes.	Updates annotations and notifies students of changes.

2.8.2 Alternative Course of Action

Actor Action	System Response
Teacher attempts to edit without existing annotations.	Displays message: "No annotations available to edit."

Exceptions:

- System database unavailable.

2.9 UC-09: View Annotated Mushaf

Identifier	UC-09
Name	View Annotated Mushaf
Purpose	Allow users to view all annotations made to the Mushaf.
Priority	Medium
Actors	Student, Teacher, Admin
Pre-conditions	User must be logged in; annotations must exist.
Post-conditions	User views annotated Mushaf with all changes displayed.
Dependencies	Annotate Mushaf, Edit Annotated Mushaf

2.9.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
User selects annotated Mushaf to view.	Retrieves and displays all annotations on the Mushaf.

2.9.2 Alternative Course of Action

Actor Action	System Response
User attempts to view Mushaf with no annotations.	Displays message: "No annotations available for this Mushaf."

Exceptions:

- System database unavailable.

2.10 UC-10: Track Student Progress

Identifier	UC-10
Name	Track Student Progress
Purpose	Allow Teachers and Students to monitor progress over time using metrics like recitation accuracy, completed verses, tajweed mastery levels, and session attendance.
Priority	High
Actors	Student, Teacher, AI Engine
Pre-conditions	Student must have prior activity data (recitations, class participation).
Post-conditions	Progress data and visualizations (charts, summaries) are displayed.
Dependencies	AI Tajweed Analysis

2.10.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Student/Teacher requests progress report.	Retrieves progress data from storage and AI Engine.
—	Generates and displays progress charts and metrics.

2.10.2 Alternative Course of Action

Actor Action	System Response
Student has no previous data.	Displays "No progress data available" and suggests initial assessment or practice tasks.
Data inconsistency detected (conflicting records).	Flags records for review and prompts admin/teacher to validate.

Exceptions:

- Database or analytics service unavailable.
- AI Engine temporarily offline (some metrics unavailable, partial view).

2.11 UC-11: Record Playback Recitation

Identifier	UC-11
Name	Record Playback Recitation
Purpose	Allow Students to record recitations (live or offline), store audio files, forward them for AI Tajweed analysis, and enable Teachers/Students to playback recordings with annotations and feedback.
Priority	High
Actors	Student, Teacher, AI Engine, Storage Service
Pre-conditions	User must be logged in; device microphone must be available and permitted.
Post-conditions	Audio file saved with metadata (verse range, timestamp); AI feedback stored (if processed); recording accessible for playback and review.
Dependencies	Login / Authenticate User, AI Tajweed Analysis, Storage Service

2.11.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Student selects Record Recitation (from session or practice area).	Displays recording UI and requests microphone permission.
Student records and stops the recording.	Saves audio file, generates metadata (reciterID, verses, duration) and stores it.
Student or teacher requests AI analysis (automatic or on-demand).	Forwards audio to AI Engine; AI returns tajweed feedback, error marks, and scores.
Teacher or student selects a saved recording to playback.	Streams audio, displays AI feedback and any teacher annotations linked to timestamps.

2.11.2 Alternative Course of Action

Actor Action	System Response
Microphone not detected or permission denied.	Shows error guidance and steps to enable microphone access.
Recording quality is poor / noisy.	Suggests re-record and may offer simple noise-check or auto-enhancement prompt.
User cancels recording mid-way.	Discards temporary file and confirms cancellation.

Exceptions:

- Storage service unavailable (notify user and retry).
- AI Engine processing timeout or failure (recording saved; analysis queued or failed with message).

2.12 UC-12: Give Recommendations

Identifier	UC-12
Name	Give Recommendations
Purpose	Provide personalized practice recommendations (verses, tajweed rules, practice exercises, session suggestions) using aggregated progress data and AI analysis, with option for teacher review/override.
Priority	Medium
Actors	Student, Teacher, AI Engine
Pre-conditions	Sufficient progress and analysis data must exist for the student.
Post-conditions	Recommendations are generated and presented; teacher may approve/modify and schedule tasks for the student.
Dependencies	Track Student Progress, AI Tajweed Analysis

2.12.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Student/Teacher requests recommendations or triggers automated recommendation cycle.	Analyzes student's progress, AI feedback, and attendance to identify weak areas.
—	Generates prioritized recommendations (target verses, tajweed rules, practice exercises, approximate practice time).
Teacher reviews recommendations (optional) and approves or modifies them.	Saves final recommendations and optionally schedules reminders/tasks in student's plan.

2.12.2 Alternative Course of Action

Actor Action	System Response
Insufficient or no data available for reliable recommendations.	Shows "Not enough data to generate recommendations" and suggests an assessment exercise.
Teacher disagrees with automated suggestion.	Teacher edits the recommendation; stores the teacher-edited version with timestamp.

Exceptions:

- Recommendation engine or analytics service unavailable.
- Data processing error (log and notify admin).

2.13 UC-13: Manage User Profile

Identifier	UC-13
Name	Manage User Profile
Purpose	Allow Students, Teachers, and Admins to view and update profile information (name, contact, photo, bio, preferences), and enable teachers to submit role-related changes (verification request).
Priority	Medium
Actors	Student, Teacher, Admin, User Verification System
Pre-conditions	User must be logged in.
Post-conditions	Profile updates are validated and persisted; role-change or verification requests are forwarded when applicable.
Dependencies	Login / Authenticate User, User Verification System

2.13.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
User selects Edit Profile.	Displays current profile details and editable fields.
User updates fields (name, email, phone, photo, bio, learning preferences).	Validates inputs, uploads profile photo to storage, and saves changes.
If Teacher requests verification or role change, user submits credentials/attachments.	Forwards documents to User Verification System and notifies admin/relevant service.

2.13.2 Alternative Course of Action

Actor Action	System Response
User provides invalid data (e.g., malformed email).	Prompts correction with specific validation message.
Photo upload fails or exceeds size limits.	Shows error and instructs user on acceptable formats/sizes.

Exceptions:

- Profile service or storage unavailable.
- Verification service down (verification requests queued).

2.14 UC-14: Verify Teacher Credentials

Identifier	UC-14
Name	Verify Teacher Credentials
Purpose	Allow Admin (or automated verification service) to review and verify teacher credentials (degrees, ijazah, sample recitations) so only authenticated/qualified teachers can teach.
Priority	High
Actors	Teacher, Admin, User Verification System, Authentication Service
Pre-conditions	Teacher must have created an account and submitted required verification documents and sample recitation(s).
Post-conditions	Teacher account status is updated to Verified (or Rejected/Needs More Info); verified teachers receive teaching privileges.
Dependencies	Register User, User Verification System, Authentication Service

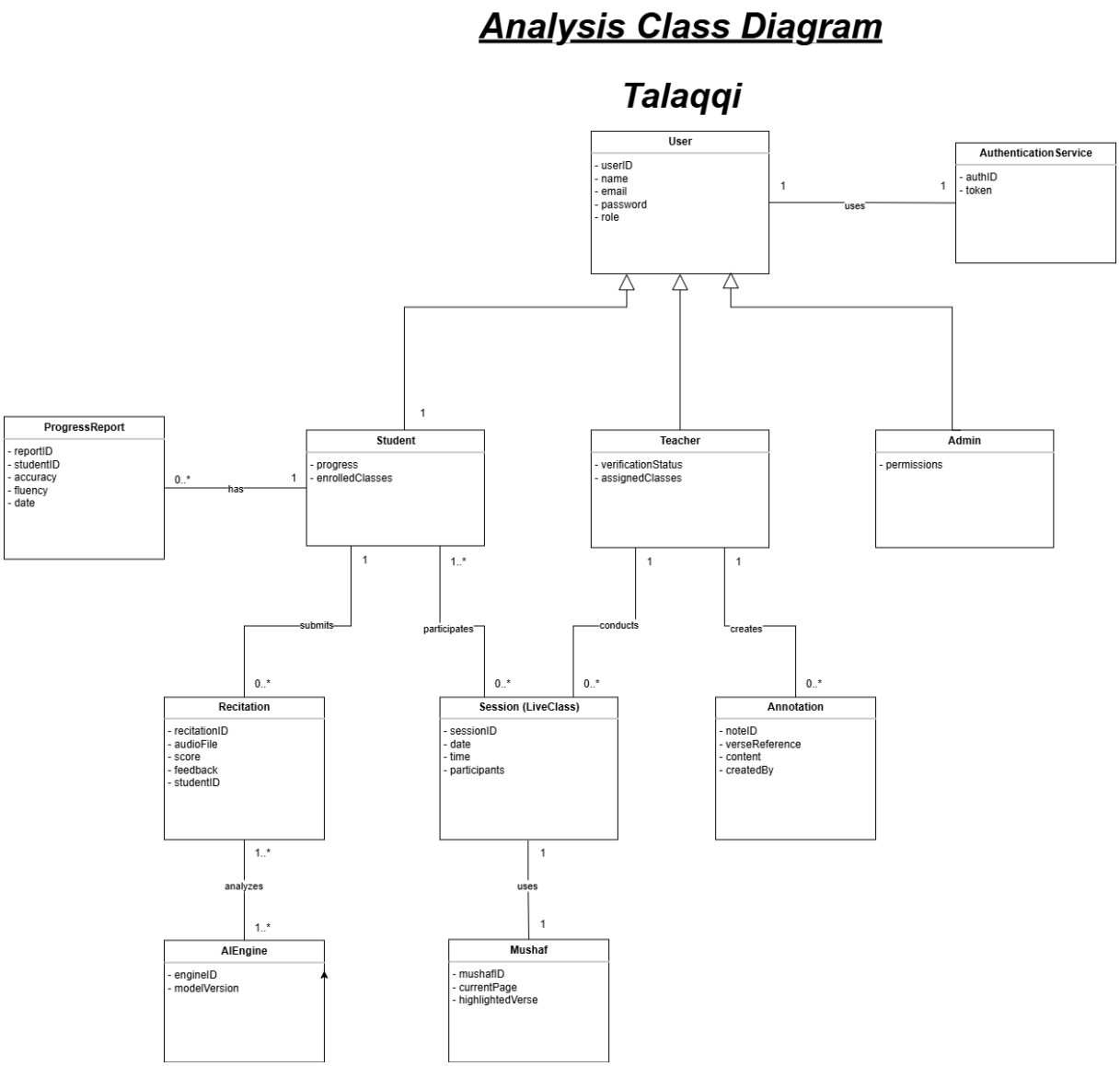
2.14.1 Typical Course of Action (Primary Flow)

Actor Action	System Response
Teacher submits verification request with documents (IDs, certificates, ijazah, sample recitation).	Stores submission and forwards to User Verification System / Authentication Service.
Admin reviews submitted documents and listens to sample recitation; may perform external checks.	Records admin decision; if approved, sets teacher status to Verified and updates permissions.
—	Notifies teacher of the decision and logs audit trail (who verified, timestamp, notes).

2.14.2 Alternative Course of Action

Actor Action	System Response
Documents are incomplete or unclear.	Requests resubmission with clear guidance and checklist.
External verification required and fails / is inconclusive.	Marks verification Rejected or Pending External Check and notifies teacher with reason and appeal steps.

Exceptions:

- Verification service or external API unavailable (requests queued).
- Security incident while handling sensitive documents (system escalates and logs incident).

3. Analysis Class Diagram